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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 3

TITLE : Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Theory

Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp3/Main1.java
Enter first number : 10
Enter second number : 20
Enter third number : 30

Sum of two : 30.0
Sum of three : 60.0
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp3/Main2.java
The roi is 8%
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp3/Exercise/Ex1.java
Enter first integer : 10
Enter second integer : 20
Enter first decimal number : 10.10
Enter second decimal number : 20.02

Method One : Sum = 30
Method two : Sum = 30.119999999999997
Number of operations : 2
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp3/Exercise/Ex2.java
----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 1
Enter the amount : Rs.1000
The total amount is = Rs.1010

----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 2
Enter the amount : Rs.1000
The total amount is = Rs.1100
```

```
----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 3
Enter the amount : Rs.1000
The total amount is = Rs.1075
```

```
----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 4
```

```
Exiting...
```

```
----- Total Orders -----
Total orders : 3
```

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q3. Method Overriding: A vehicle tracking system needs to calculate the maximum speed of different types of vehicles.

- A generic vehicle has a base speed.
- A car has a higher max speed.
- A bike has an even higher max speed.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp3/Exercise/Ex3.java
The speed for the vehicle is : 100 km\hr
The speed for the vehicle is : 200 km\hr
The speed for the vehicle is : 300 km\hr
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q4. Method Overloading: Payment Processing System

A company is building a payment gateway that allows users to pay using different methods:

- Credit Card (card number & CVV)
- Bank Transfer (account number & IFSC code)
- UPI (UPI ID)

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp3/Exercise/Ex4.java
----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 1
Paid using credit card.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 2
Paid using Bank Transfer.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 3
Paid using UPI ID.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 4

Exiting...
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This assignment helped in understanding the concepts of method overloading and static variables and methods in Java. It demonstrated how method overloading enables multiple methods with the same name to perform different tasks based on their parameter lists, improving code flexibility and readability. The use of static variables and methods showed how data and functionality can be shared among all objects of a class, reducing memory usage and simplifying program design. These concepts are fundamental to object-oriented programming and are widely used in developing efficient, reusable, and well-structured Java applications.